■■ COMPLETE ENGINEERING, COMPONENT MASTERY, & FULL ENGINEER NETWORK DIRECTORY

This comprehensive handbook details the absolute mechanical fundamentals of ship modification, component statistics, experimental vendor items, and the full, complete galactic network of all 20 human-occupied space engineers.

1. The Foundational Engineering Protocol

Engineering represents the ultimate progression track in Elite Dangerous. Working with any engineer requires completing a strict three-stage pipeline sequence: **Discovery** (learning about their existence), **Invitation** (meeting their unique system tracking conditions), and **Levelling** (donating a rare resource commodity to permanently open their workshop tiers).

■ INTERACTIVE FLIGHT TRAINING RESOURCE:
To visually trace the initial materials gathering loops and structural mechanics, refer to the Elite Dangerous Engineering Guide: Step-by-Step Tutorial on YouTube.

Complete Galactic Engineer Registry (All 20 Bubble Engineers)

Below is the complete blueprint breakdown of all 20 human-space engineers, categorized by specialization matrix alongside exact unlock metrics and component capabilities:

Engineer & System	Primary Capabilities	Unlock Conditions & Donation Requirements
Felicity Farseer Deciat System	• **Frame Shift Drive (G5)** • Thrusters (G3) • Sensors (G3)	**Discovery:** Hit Exploration rank *Scout* or higher. **Donation:** Provide 1 unit of Meta-Alloy.
Elvira Martuuk Khun System	• **Frame Shift Drive (G5)** • Shield Generators (G2) • Thrusters (G2)	**Discovery:** Travel 300+ Light Years away from starting port. **Donation:** Provide 1 unit of Soontill Relics.
Professor Palin Arque System	• **Thrusters (G5)** • Frame Shift Drive (G3)	**Discovery:** Travel 5,000+ Light Years from your origin system. **Donation:** Provide 25 units of Sensor Fragments.
Chloe Sedesi Shenve System	• **Thrusters (G5)** • Frame Shift Drive (G3)	**Discovery:** Learn from Elvira Martuuk (G3+ proficiency). **Donation:** Provide 25 units of Sensor Fragments.
Marco Qwent Sirius System	• **Power Plants (G4)** • Power Distributors (G3)	**Discovery:** Learn from Elvira Martuuk (G3). Requires *Sirius System Permit*. **Donation:** Provide 25 Modular Terminals.
Bill Turner Alioth System	• **Plasma Accelerators (G5)** • Sensors / Life Support (G3)	**Discovery:** Learn from Marco Qwent (G3). Requires *Alioth System Permit*. **Donation:** Provide 50 units of Bromellite.
The Dweller Wyrd System	• **Power Distributors (G5)** • Pulse & Burst Lasers (G4)	**Discovery:** Transact at 5+ distinct Black Markets. **Donation:** Pay a fee of 500,000 Credits.
Lei Cheung Laksak System	• **Shield Generators (G5)** • Sensors (G5)	**Discovery:** Trade commodities at 50+ unique stations. **Donation:** Deliver 200 units of Gold.

Didi Vatermann Leesti System	• **Shield Boosters (G5)** • Shield Generators (G3)	**Discovery:** Learn from Lei Cheung. Reach Trade rank *Merchant*. **Donation:** Deliver 50 units of Lavian Brandy.
Chesterfield Flandry Castellan System	• **Shield Boosters (G5)** • Shield Generators (G3)	**Discovery:** Requires Powerplay 2.0 system alignment matrices. **Donation:** Provide specialized operational intelligence logs.
Liz Ryder Eurybia System	• **Missile Racks & Torpedoes (G5)** • Armor Reinforcements (G1)	**Discovery:** Available globally. **Donation:** Deliver 200 units of Landmines.
Hera Tani Kuwemani System	• **Power Plants (G5)** • Surface Scanners (G5)	**Discovery:** Learn from Liz Ryder. Imperial Rank *Outsider*. **Donation:** Provide 50 units of Kamitra Cigars.
Juri Ishmaak Giryak System	• **Sensors (G5)** • Surface Scanners (G5)	**Discovery:** Learn from Felicity Farseer. Complete 50+ Combat Bonds. **Donation:** Cash in 1,000,000 Credits of Federal Combat Bonds.
Tod 'The Blaster' McQuinn Wolf 397 System	• **Multi-cannons (G5)** • Railguns (G2)	**Discovery:** Earn cumulative local bounty vouchers. **Donation:** Cash in 100,000 Credits of local Bounty Vouchers.
Selene Jean Kukawa System	• **Hull Reinforcements (G5)** • Armor Modifications (G5)	**Discovery:** Learn from Tod McQuinn. Mine 500+ tons of raw ore. **Donation:** Provide 50 units of Painite ore.
Broo Tarquin Muang System	• **Beam, Pulse, Burst Lasers (G5)**	**Discovery:** Learn from Hera Tani. Combat rank *Competent* or higher. **Donation:** Provide 50 units of Fujin Tea.
Tiana Fortune Achenar System	• **Limpet Controllers (G5)** • Scanners & Sensors (G5)	**Discovery:** Learn from Liz Ryder. Exploration rank *Surveyor*. **Donation:** Deliver 50 Decoded Emission Data logs.
Zachariah Nemo Yoru System	• **Fragment Cannons (G5)** • Multi-cannons (G2)	**Discovery:** Learn from Tod McQuinn. Allied with Party of Yoru. **Donation:** Provide 25 units of Xanit ore.
Ram Tah Meene System	• **Chaff / Heat Sinks / Point Def (G5)**	**Discovery:** Unlocks via baseline utility mod tracking telemetry. **Donation:** Provide 50 Classified Scan Databanks.
Lori Jameson Shinrarta Dezhra	• **Sensors / Life Support / Scoops (G5)**	**Discovery:** Learn from Marco Qwent. Combat rank *Dangerous*. **Donation:** Deliver 25 units of Kongga Ale.

2. Understanding Component Classes & Performance Ratings

Every module inside an outfit array relies on a rigid classification matrix: a **Size Class (Numeric 1–8)** and a **Performance Rating (Alphabetical A–E)**. Installing conflicting component ratings will instantly compromise structural mechanics.

Rating	Functional Designation & Performance Parameters
A-Rated	**Peak Performance:** Offers highest power outputs, optimal thermal ventilation thresholds, and top-tier capabilities. Standard for high-intensity combat configurations.
B-Rated	**Armored Heavyweight:** Possesses maximum module health points to survive internal subsystem breach drops, but heavily penalizes speed/range due to massive weight.
C-Rated	**Cost-Efficient:** Baseline performance balancing intermediate output fields at highly accessible market purchase points.
D-Rated	**Lightweight Fleet:** Lowest physical mass metrics in the galaxy. **Mandatory configuration for Exploration and Exobiology loops** to maximize unladen jump distances.
E-Rated	**Standard Issue:** Factory-default modules deployed on stock ships. Extremely inefficient thermal parameters and high weight footprints. Replace immediately.

3. Experimental Vendor Modules: Supercruise Overcharge (SCO)

Commanders can bypass the standard engineering loops completely for certain mechanics by purchasing pre-modified experimental gear straight from commodity markets using credits.

- **The Supercruise Overcharge (SCO) FSD:** Explicitly designated with an **'SCO'** identifier within outfitting arrays. It functions like a high-grade Frame Shift Drive but enables pilots to hit their *Engine Boost* controls while cruising in planetary supercruise. This triggers immense acceleration vectors, allowing ships to cross thousands of light-seconds to distant planets in moments.
- **Procurement Vector:** These are classified as C-rated out of the box but boast base jump fields equivalent to A-rated models. Use **INARA.cz ➔ Outfitting** to scan nearby stations stocking specialized SCO variants tailored to your hardpoint hull sizes.

■ INTERACTIVE FLIGHT TRAINING RESOURCE:
To observe a live flight alignment demonstrating heat vent management during overcharge propulsion, watch the How to Find & Use Supercruise Overcharge (SCO) Drive on YouTube.

4. Guardian Alien Technology Overviews

Guardian Tech parts are powerful hybrid components reverse-engineered from extinct alien data logs. **Unlike human engineering, they are unlocked exactly once at a Technology Broker and can then be purchased with standard credits infinitely.**

- **Guardian FSD Booster:** An optional internal module that injects a flat, un-engineerable bonus of up to **+10.5 Light Years** of jump capability straight into any ship hull.
- **Guardian Gauss Cannon:** High-velocity, absolute armor-piercing kinetic rails. This represents the undisputed meta hardpoint choice for sniping internal hearts during high-intensity **Anti-Xeno (AX) Thargoid combat**.

■ INTERACTIVE FLIGHT TRAINING RESOURCE:
For a complete walkthrough showing pylon activation algorithms and ancient relic processing steps at alien sites, review the How to Unlock the Guardian FSD Booster in Elite Dangerous on YouTube.

5. Experimental Effects & The Pinned Blueprint System

Modification blueprints permanently overhaul a module's core stats across five distinct ranks, while **Experimental Effects** act as a single final modification slot to highly specialize performance parameters.

- **Dirty Thrusters + Drag Drives:** The undisputed standard for combat hull agility, heavily multiplying maximum cruising velocity and turn rates.
- **Increased Range FSD + Mass Manager:** The primary configuration framework for long-range navigation, extracting maximum light-year output.
- **The Pinning Strategy:** You can select one blueprint per engineer and select **'Pin Blueprint'**. This allows you to remotely execute that modification sequence grade progression at *any space station hangar in the universe* via the Remote Engineering tab, avoiding unnecessary physical travel across the bubble. *Note: Adding final Experimental Effects still requires a physical flight to the engineer's planet base.*

6. The Third-Party Engineering Tool Suite

Do not attempt to navigate resource farming grids blindly. Keep these specialized applications open on an auxiliary monitor to automate your optimization pipelines:

- **INARA.cz (The Material Inventory Ledger):** Sync your profile logs via the EDMC tool background client. Pin blueprints on the web portal to generate an accurate shopping list. INARA reads your inventory logs and identifies the closest **Material Trader** to swap your overstocked items.
- **Coriolis.io / EDSY (Ship Sandbox Simulators):** Virtual garage sandboxes allowing you to pre-configure hull setups, apply theoretical engineering grades, and evaluate power consumption fields before spending a single credit in-game.
- **ED Odyssey Materials Helper (GitHub App):** A background executable tracking client journal text lines. It traces recipes and presents a desktop overlay detailing precisely what signals, wreckage nodes, or settlements to farm.

■ INTERACTIVE FLIGHT TRAINING RESOURCE:
To learn how to execute high-grade emissions material farm paths and optimize trading returns at market hubs, watch the Elite Dangerous Materials Gathering Tutorial on YouTube.

■■ CRITICAL FLIGHT OPERATIONS & THERMAL MANAGEMENT WARNING:
Engineering modifications heavily warp stock module tolerances, frequently shifting power grid draws and heat generation models. Furthermore, **deploying the Supercruise Overcharge (SCO) engine boost will cook your vessel rapidly if sustained.** The SCO function drains fuel reservoirs at an astronomical rate and causes internal ship heat to spike exponentially past **100%** within seconds, resulting in immediate structural subsystem failure. Always verify your ship's peak thermal limits on Coriolis, carry active Heat Sink Launchers, and limit SCO engagement to short, rapid bursts.